

8) . First we check well definedness

$$\text{Let } v+W = v'+W \quad \& \quad u+W = u'+W$$

$$\Rightarrow v-v' \in W \quad \& \quad u-u' \in W$$

$$\Rightarrow (v-v') + (u-u') \in W$$

$$\Rightarrow (v+u) - (v'+u') \in W$$

$$\Rightarrow (v+u)+W = (v'+u')+W$$

$$\text{Let } v+W = v'+W$$

$$\Rightarrow v-v' \in W$$

$$\Rightarrow a \cdot (v-v') \in W$$

$$\Rightarrow a \cdot v - a \cdot v' \in W$$

$$\Rightarrow a \cdot v + W = a \cdot v' + W$$

Thus V/W has well defined addition & scalar multiplication

Easy to verify vector space axioms because they are true for V . for example

$$\begin{aligned}
 (a+b) \cdot (v+w) &= (a+b) \cdot v + w \\
 &= (av + bv) + w \\
 &= (av + w) + (bv + w) \\
 &= a \cdot (v+w) + b \cdot (v+w)
 \end{aligned}$$

Q2. Basis of $U = \{ (2, -1, 0, 1, 0), (0, 1, 0, 1, 3, -1) \}$

Extend this to basis of $\mathbb{R}^4$.

$(1, 0, 1, 0, 1, 0)$, $(0, 1, 0, 1, 1, 0)$ work.

so basis of

$$\mathbb{R}^4/U = \{ (1, 0, 1, 0, 1, 0) + U, (0, 1, 0, 1, 1, 0) + U \}$$

Q3. We know

$$\dim(U+W) = \dim(U) + \dim(W) - \dim(U \cap W)$$

We have

$$\dim(U) = \dim(W) = 4$$

$$\text{So } \dim(U+W) = 8 - \dim(U \cap W) \quad (1)$$

$$\max\{\dim(U), \dim(W)\} \leq \dim(U+W) \leq \dim(\mathbb{R}^6)$$

$$4 \leq \dim(U+W) \leq 6 \quad (2)$$

By (1) & (2)

$$2 \leq \dim(U \cap W) \leq 4$$

So $U \cap W$ has basis of length at least two.
So there exist two vectors in $U \cap W$ such that neither is scalar multiple of other.

Q4. For $U \cap W \Rightarrow$

Let $v \in U \cap W$

$$\Rightarrow v = a(1, 1, 0, -1) + b(0, 1, 3, 1)$$

also $v = x(0, -1, -2, 1) + y(1, 2, 2, -2)$

$$\Rightarrow a(1, 1, 0, -1) + b(0, 1, 3, 1) = x(0, -1, -2, 1) + y(1, 2, 2, -2)$$

solve

$$x=1, y=1, a=1, b=0$$

$$\Rightarrow v = (1, 1, 0, -1)$$

$$U \cap W = \text{span} \{ (1, 1, 0, -1) \}$$

$$\dim(U+W) = 2+2-1=3$$

Basis of $U+W$ = Extend basis of U to $U+W$

check if $(0, -1, -2, 1)$ is in U
If not then

$$\text{Basis of } U+W = \{ (1, 1, 0, -1), (0, 1, 3, 1), (0, -1, -2, 1) \}$$

Q5.

$$\text{A basis of } U = \{ 1, (x-2)(x-5)(x-6), (x-2)(x-5)(x-6)^2 \}$$

Extend U to basis of P_4

$$\{ 1, x, x^2, (x-2)(x-5)(x-6), (x-2)(x-5)(x-6)^2 \}$$

$$\text{Take } W = \text{span} \{ x, x^2 \}$$

Then $P_4 = V \oplus W$

Q6. b. Counter example :-

$$V = \mathbb{R}^2$$

$$U_1 = \{ (x, 0) \mid x \in \mathbb{R} \}$$

$$U_2 = \{ (x, x) \mid x \in \mathbb{R} \}$$

$$U_3 = \{ (0, x) \mid x \in \mathbb{R} \}$$

$$\dim(U_1 + U_2 + U_3) = 2$$

$$\dim(U_1) = \dim(U_2) = \dim(U_3) = 1$$

$$\dim(U_1 \cap U_2) = \dim(U_2 \cap U_3) = 1$$

$$\dim(U_1 \cap U_3) = \dim(U_1 \cap U_2 \cap U_3) = 0$$

$$2 \neq 1 + 1 + 1 - 1 - 1 - 0 + 0$$

a. Let $v_1, v_2, \dots, v_m$ be basis of $U_1 \cap U_2$
then $\dim(U_1 \cap U_2) = m$.

Extend this to basis of U_1

say $v_1, v_2, \dots, v_m, u_1, u_2, \dots, u_j$

$$\text{then } \dim(U_1) = m + j$$

Also extend $v_1, v_2, \dots, v_m$ to basis of U_2

say $v_1, v_2, \dots, v_m, w_1, w_2, \dots, w_k$

$$\text{then } \dim(U_2) = m + k$$

Now we show $v_1, \dots, v_m, u_1, \dots, u_j, w_1, \dots, w_k$ is basis of $U_1 + U_2$, because then

$$\begin{aligned} \dim(U_1 + U_2) &= m + j + k \\ &= (m + j) + (m + k) - m \end{aligned}$$

$$\text{clearly } \{v_1, v_2, \dots, v_m, u_1, \dots, u_j, w_1, \dots, w_k\} \subseteq U_1 \cup U_2 \\ \subseteq U_1 + U_2$$

The span of list contains $U_1 \cup U_2$ & hence equal to $U_1 + U_2$. Thus we need to show now linear independence.

Suppose

$$a_1 v_1 + \dots + a_m v_m + b_1 u_1 + \dots + b_j u_j + c_1 w_1 + \dots + c_k w_k = 0 \\ \Rightarrow$$

$$c_1 w_1 + \dots + c_k w_k = -a_1 v_1 - \dots - a_m v_m - b_1 u_1 - \dots - b_j u_j$$

$$\Rightarrow \text{L.H.S.} \in U_1$$

All the w_i 's are in U_2 so

$$\text{L.H.S.} \in U_1 \cap U_2$$

$$\Rightarrow c_1 w_1 + \dots + c_k w_k = d_1 v_1 + \dots + d_m v_m$$

But $\{v_1, \dots, v_m, w_1, \dots, w_k\}$ is l.i. so

$$c_i = 0, d_i = 0$$

$$\Rightarrow a_1 v_1 + \dots + a_m v_m + b_1 u_1 + \dots + b_j u_j = 0$$

But $\{v_1, \dots, v_m, u_1, \dots, u_j\}$ is l.i.

$$\text{so } a_i = 0, b_j = 0$$